Chapter 17 exercises • Distributed systems

OS17-E01 • Speedup and fault domains

A 120 s job contains30 s serial and90 s evenly parallel across six nodes, plus6 s coordination. Three independent interchangeable read replicas each fail with probability.02.

1. Compute T6, speedup, and the infinite-node bound ignoring coordination.
2. With reliable networking/metadata and any replica sufficient, find total failure probability. What if all share a rack?

OS17-E02 • Moving data or computation

Data 512 MiB, result2 MiB, throughput64 MiB/s, RTT10 ms, remote computation200 ms. Ignore request, queue, and other costs.

1. Compare full-data transfer with remote computation/result costs.
2. Is sending code migration of a running process?

OS17-E03 • BDP and windows

Link200 Mbit/s, RTT50 ms, outstanding limit128 KiB, no loss/other bottleneck.

1. Find BDP and window throughput bound.
2. Does upgrading the link to1 Gbit/s multiply that window bound by five?

OS17-E04 • DNS cache and identity

At0, store.lab.test resolves to198.51.100.20 with TTL300 s. Authority changes to.30 at 100.

1. What can be observed at 150 and after 300 expiry? Must every label have a server?
2. Are domain, IP, port equivalent to user, PID, or permanent host identity?

OS17-E05 • Routes and packet pipelines

Graph:A–C2, C–B2, A–D1, D–B5. Separately, two1 Mbit/s store-and-forward links carry four1000 B packets without headers, propagation, queueing.

1. Trace tentative B distances and the route after C–B fails.
2. Compare completion time with one4000 B message.

Chapter 17 exercises • Distributed systems (continued)

OS17-E06 • IPv4, ARP, and a short frame

Use the chapter's no-NAT two-LAN A/R/B model. TCP/IPv4 headers are20 B each;payload 4 B.

1. Whom does A ARP? Give both hops' MAC and IP endpoints.
2. Find padding, frame length, and wire occupancy including the specified preamble/IFG.

OS17-E07 • Indistinguishable failure observations

A receives no ping reply for 200 ms;network/scheduling delay is unbounded.

1. Give equal A histories with different B liveness.
2. WithH=.5 s, send delay≤.1 s, one-way delay≤.05 s, no loss, find maximum heartbeat arrival gap.

OS17-E08 • Fencing and retries

A holds epoch42, B will receive43. Counter 10 handles increment(id17), whose reply may be lost.

1. Why is timeout-based self-election unsafe? State resource-side conditions.
2. How does retry preserve11 rather than increment to12? Identify crash windows.

OS17-E09 • DFS naming and cache cost

Path→FID(vol9, file 2, g4), vol9→serverA. Of100 reads,20 miss at 6 ms and80 hit at.2 ms. Each miss transfers4 KiB.

1. Which map changes on movement to serverB? What about reusing file 2?
2. Compute total delay/payload traffic, then add1 ms validation per read.

OS17-E10 • Consistency and partitions

C1 cachesv0 at 0, C2 completesv1 at 2, C1 next validates at 8. Current time3.

1. Can periodic validation returnv0? Is immediate new-version visibility provided?
2. What premise is needed during partition for callback/lease writes to finish before all new reads excludev0?

Chapter 17 exercises • Distributed systems (continued)

OS17-E11 • NFSv3 verifier changes

Client WRITE D returnsUNSTABLE, verifierX. Server restarts before COMMIT, loses volatile D, and usesY.

1. What follows from COMMIT returningY≠X? When may D be discarded?
2. Does stateless mean no cache or every WRITE synchronously persists?

OS17-E12 • Historical HDFS and failover capacity

Hadoop1.2.1, file 150 MiB, custom64 MiB blocks, RF3. Two servers each handle100 requests/s, total load150/s.

1. Give block payloads, total replica payload, and NameNode/data paths.
2. Can one surviving server sustain the same load?

References and attribution

Abraham Silberschatz, Peter B. Galvin, Greg Gagne. Operating System Concepts, 9th Edition.

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https://github.com/xinhi6716/computer-science-self-study

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